Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 10 - Week 10 (March 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1

Which of the following denote properties of Hashed Timelock Contracts (HTLC)?

a. If the secret is not revealed, the funds are permanently locked and cannot be recovered.
b. If the secret is not revealed, all contracts using that secret are automatically executed.
c. HTLCs restrict the spending of funds until a valid secret (preimage) is revealed or a predefined timeout occurs.
d. If the secret is not revealed, the funds are automatically transferred to the recipient after the timeout.

Answer: (c)

Detailed solution:
An HTLC works using two simple rules. First, the receiver can take the funds only if they reveal the correct secret that matches a given hash. Second, if they don't reveal it within a fixed time, the sender gets the money back. This way, either the receiver claims the funds with the secret, or the sender safely recovers them after the timeout, so no one loses money unfairly.

QUESTION 2

If a participant in an atomic cross-chain swap halts during the "triggering phase" (after the hashlock secret has been revealed), what is the resulting state for a conforming counterparty?

a. The counterparty is "worse off" because they have lost their asset.
b. The protocol guarantees that the conforming party either receives their intended asset or retains their original asset.
c. All assets on all involved blockchains are permanently frozen.
d. The counterparty must wait for a consensus-level view change to reclaim their funds.

Answer: (b)

Detailed solution:
Please refer to week 10 lecture 47.

QUESTION 3

Which of the following are inherent risks or disadvantages associated with using a Centralized Trusted Third Party (TTP) for cross-chain asset transfers?

a. Users must relinquish control of their funds to the third party.
b. The third party represents a single point of failure.
c. Cross-chain transfers cannot be performed using a third party.
d. Users may lose funds if the third party is compromised or shuts down.

Answer: (a), (b), and (d)

Detailed solution:
Using a centralized TTP (like an exchange) means you have to give up control of your funds and trust the third party. This creates a single point of failure—if the exchange gets hacked or shuts down suddenly, you can lose your funds. Option (c) is incorrect because TTPs are actually used to enable transfers across different protocols, not prevent them.

QUESTION 4

Why is "Consensus-driven" data transfer emphasized for permissioned blockchains?

a. To ensure that only the administrator of the network can authorize a transfer.
b. To guarantee that the data being moved is not a temporary or unconfirmed state of the source ledger.
c. Because permissioned blockchains do not support hashlocks.
d. To allow the destination network to participate in the mining process of the source network.

Answer: (b)

Detailed solution:
In permissioned systems (like Hyperledger Fabric), data transfer must be "verifiable." We use consensus-driven data to ensure that the information being sent from the source has been finalized by the required number of peers. This prevents the destination network from acting on data that might be "rolled back" or "orphaned."

QUESTION 5

In the three-party atomic swap scenario (Alice, Bob, Carol) involving an alt-coin, Bitcoin, and a car title, why is the sequence of timelocks (6Δ, 5Δ, 4Δ) critical?

a. To ensure the car title is transferred before Alice spends her alt-coins.
b. To prevent the Bitcoin network from crashing due to high transaction volume.
c. To allow Carol to claim her Bitcoin and Bob to claim his alt-coins before their respective contracts expire, even if the secret s is revealed late in the chain.
d. To ensure that Alice, as the generator of the secret, has the least amount of time to act.

Answer: (c)

Detailed solution:
In a multi-party swap, the timelocks must be "staggered" (e.g., 6Δ, 5Δ, 4Δ). This gives each person in the chain enough of a buffer to react. If Carol uses the secret to claim her Bitcoin at 4Δ, Bob still has until 5Δ to use that same secret to claim his alt-coins from Alice.

QUESTION 6

According to the permissioned blockchain interoperability (e.g., between TradeLens and We.Trade), what must be included with the data moving from the source network to the destination?

a. A set of endorsements or signatures representing the consensus view of the source network.
b. A proof that the data has been encrypted using the destination network's private key.
c. Verification that the data sharing complies with the source network's "Exposure Control" policy.
d. A physical bill of lading scanned into a PDF format.

Answer: (a) and (c)

Detailed solution:
Please also refer to week 10 lecture 48.

QUESTION 7

During the Hyperledger Indy identity configuration, which statements correctly describe the roles of Stewards and Trust Anchors?

a. Stewards have the permission to register new DIDs (verinyms) on the ledger.
b. Trust Anchors are responsible for maintaining the RBFT consensus nodes.
c. Trust Anchors, such as a University, have the authority to issue verifiable credentials to users.
d. Stewards are pre-approved participants who can approve the transition of a user to a Trust Anchor role.

Answer: (a), (c), and (d)

Detailed solution:
In Hyperledger Indy, **Stewards** are the primary authorities who can write to the ledger and approve **Trust Anchors**. Trust Anchors (like a University) then have the power to issue credentials to users. Option (b) is wrong because Stewards usually run the validator nodes (RBFT nodes), not Trust Anchors.

QUESTION 8

In the permissioned interoperability relay architecture, what does the "Data Acceptance Policy" in the destination network (Network A) evaluate?

a. The current exchange rate of the assets being transferred.
b. Whether the received data has a sufficient number of valid attestations/signatures from the source network.
c. The identity of the individual programmer who wrote the source chaincode.
d. If the source network (Network B) has a higher market cap than Network A.

Answer: (b)

Detailed solution:
The Data Acceptance Policy is a gatekeeper for the destination network. It defines the "trust threshold" specifically: how many signatures or endorsements from the source network are required before the destination network considers the incoming data "valid" and writes it to its own ledger.

QUESTION 9

Why does Hyperledger Indy use a "Genesis Transaction" file (pool_transactions_genesis) during the application setup?

a. It contains the history of every identity ever created since the start of the web.
b. It is a legal document that Alice must sign before using the system.
c. It reboots the nodes every time a new transaction is made.
d. It provides the initial IP addresses, ports, and keys of the nodes in the Indy pool so the SDK can connect to the network.

Answer: (d)

Detailed solution:
The Genesis Transaction file is the starting point (the "bootstrap"). It tells the client application exactly where the nodes are located on the internet and what their public keys are. Without this file, the Indy SDK wouldn't know which servers to talk to.

QUESTION 10

In the Indy tutorial, the University sets `support_revocation: False` when creating a Credential Definition. What is the implication of this setting?

a. The University can never issue more than one credential.
b. Once issued, the University cannot use an Indy Revocation List to invalidate that specific transcript in the future.
c. Alice is forbidden from showing her transcript to any company.
d. The credential will automatically delete itself from Alice's wallet after 24 hours.

Answer: (b)

Detailed solution:
Please also refer to week 10 lecture 50.